Validity Analysis: INJONGO

Target context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Overall risk: **HIGH**

Deployment Context

Use case: Amharic intercity-travel booking bots for Sky Bus, Selam Bus, and Ride covering routes, fares, and seat changes

Target population: Inter-regional commuters and travelers booking long-distance buses and ride-hail trips via smartphone apps

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	2	Significant gaps	high
Input Content ⚠	3	Moderate gaps	high
Input Form ✓	5	Strong alignment	high
Output Ontology	2	Significant gaps	high
Output Content ⚠	2	Significant gaps	high
Output Form ✓	4	Minor gaps	high
Average	**3.0**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

INJONGO is the most relevant existing benchmark for Amharic intent detection and slot-filling, and the only large-scale resource of its kind. For the Ethiopian intercity/ride-hail deployment, it provides genuine value in three respects: (1) Amharic is included as a first-class language with culturally grounded entities (Ethiopian banks, TeleBirr, Ethiopian cities including Dire Dawa, Bahir Dar, Hawassa, Arba Minch, and even Ge'ez calendar months Meskerem and Hamle as DATE slot values); (2) text/Ge'ez script and structured intent+slot output exactly match deployment requirements; (3) annotation quality after review is high (κ=0.933 for Amharic). However, three substantive gaps limit construct validity for this specific deployment: (a) the 40-intent travel taxonomy is air-travel-framed and entirely lacks intercity bus, seat-class, fare-tier, and operator-specific rebooking intents — confirmed empirically in the Amharic split (HIGH-priority IO concern); (b) Ethiopian Birr is absent as a

CURRENCY/MONEY slot value in the sampled Amharic data while other tracks' currencies (Naira, Rand) are well-represented, and most Amharic DATE slots use relative expressions rather than named Ge'ez months (HIGH-priority IC residuals); (c) no code-switched Amharic–Oromo / Amharic–Tigrinya utterances exist despite the L2-heavy multi-regional user base, and annotator regional demographics are undisclosed (HIGH-priority OC concern). The benchmark is suitable as a foundation but requires deployment-specific supplementation before being treated as sufficient evaluation evidence.

Practical Guidance

What This Benchmark Measures

INJONGO measures Amharic NLU performance on a 40-intent / 23-slot taxonomy spanning banking, home, kitchen-and-dining, travel, and utility domains, using accuracy (intent) and F1 (slot). It provides a valid measure of (i) general Amharic text comprehension by encoder, encoder-decoder, and LLM-prompted models, (ii) culturally grounded slot extraction for Ethiopian banks, payment platforms (TeleBirr), and major intercity cities, and (iii) cross-lingual transfer behavior from English to Amharic. Strongest dimensions for this deployment are input form (full text/Ge'ez match) and output form (structured intent+slot output matches NLU module). It can support model selection across encoder, decoder, and LLM families for the Amharic NLU layer.

Construct Depth

Construct depth is shallow-to-moderate for the deployment's actual task. The benchmark probes generic conversational NLU on Amharic and partially covers Ethiopian cultural grounding in banking and selected travel slots, but it does not probe (a) intercity bus operator booking flows, (b) seat-class/fare-tier reasoning, (c) Ethiopian Birr fare extraction in travel context, (d) Ethiopian-clock-convention TIME extraction, (e) Ge'ez-calendar dense temporal reasoning beyond the few sampled instances, or (f) code-switched Amharic–Oromo / Amharic–Tigrinya robustness. High benchmark scores would not justify confidence that a deployed model handles fare extraction, holiday-surge bookings, or L2-speaker code-switched utterances correctly.

What Else You Need

(1) For input ontology: build an operator-specific intent inventory (Sky Bus, Selam Bus, Ride) covering seat-class upgrade, road-closure rebooking, holiday-surge booking, and operator-specific cancellation; benchmark on this. (2) For input content: collect or synthesize an Amharic test slice with Ethiopian Birr-denominated fares, dense Ge'ez calendar month references including Pagume, Ethiopian-clock-convention TIME values, and the full set of intercity corridor toponyms; supplement with code-switched Amharic–Oromo and Amharic–Tigrinya utterances. (3) For output content: re-annotate or expert-validate a slice using non-Addis-centric L2 Amharic annotators, including rural and Tigray/Oromia-region speakers, to assess label validity for code-switched utterances. (4) Supplement with EthioBenchmark NER/POS as auxiliary evidence, and operator-log-derived utterances for ecological validity.